

Table of Contents

<u>0.2-1 Project description</u>	<u>1</u>
<u>0.2-2 Design input</u>	<u>1</u>
<u>0.2-3 Design Results</u>	<u>2</u>



0.2-1 | Project description

Design of embedded pole foundation for the flagpole design example in Appendix A of NAAMM/FP 1001-07, “Guide Specifications for Design of Metal Flagpoles” Embedded pole foundation design is per 2024 IBC Eq 6-1 and Table 18-I-A. Soil properities are per Table 1806.2 in the 2024 IBC.

0.2-2 | Design input

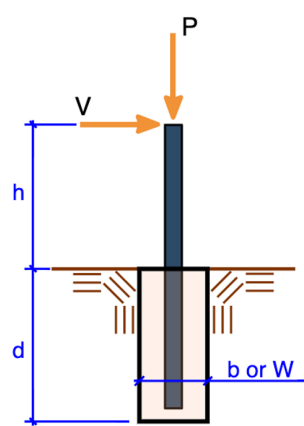


Fig. 2.1 - Calculation Diagram

Table 2.1: Design input

variable	value	[value]	description
Mbase	24.84 ft-kip	33.67 mkN	moment at base of flagpole
P	24.84 kips	110.47 kN	horizontal load
b	24.00 inch	60.96 cm	width of concrete drilled pier
PFP	200.00 p_cf	31.42 kN_m3	allowable lateral bearing pressure - sandy gravel
h	1.50 ft	0.46 m	height
d	10.00 ft	3.05 m	initial guess for embedment depth

Table 2.2: Load embedment functions (pole_embed.py)

Function	Docstring
Depth_nonconstrained(d, P, h, dia, PFP, tol)	Calculate required pole embedment using Eq. 18-1
Depth_constrained(d, P, h, dia, PFP)	Calculate required pole embedment using Eq. 18-2

0.2-3 | Design Results

Eq. 3.2: Required embed - nonconstrained

```
depth1 = Depth_nonconstrained(d, P, h, ↵
↵ b, PFP, 2)
```

depth₁ = 15.52 ft [depth₁] = 4.73 m | Required embed - nonconstrained

b	h	P	d	PFP
24.00 inch	1.50 ft	24.84 kips	10.00 ft	200.00 p_cf
width of concrete drilled pier	height	horizontal load	initial guess for embedment depth	allowable lateral bearing pressure - sandy gravel
-	-	-	-	-

Eq. 3.3: Required embed - constrained

```
depth2 = Depth_constrained(d, P, h, b, ↵
↵ PFP)
```

depth₂ = 7.34 ft [depth₂] = 2.24 m | Required embed - constrained

b	h	P	d	PFP
24.00 inch	1.50 ft	24.84 kips	10.00 ft	200.00 p_cf
width of concrete drilled pier	height	horizontal load	initial guess for embedment depth	allowable lateral bearing pressure - sandy gravel
-	-	-	-	-

